The Information Profit Principle:

Derivation for Reflexive Systems and Empirical Validation Across Self-Maintaining Systems

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2026

Preprint.
Archival version (Zenodo): https://doi.org/10.5281/zenodo.20170102
UGP series hub (Zenodo): https://doi.org/10.5281/zenodo.20168144

Abstract

A reflexive self-maintaining system must generate information structure faster than it dissipates it. Within the MFRR framework, this requirement yields a sharp dimensionless threshold, the *Information Profit Threshold* (IPT):

$$\mathrm{IPT} = 1 + \tfrac{\Lambda}{2}, \quad \Lambda = \frac{\ln \varphi}{\ln 2\pi} \approx 0.2618,$$

where $\varphi = (1+\sqrt{5})/2$, giving IPT ≈ 1.1309. A system with generative-to-drain ratio $\rho = G/D >$ IPT can sustain coherent self-referential processing; one with $\rho \leq$ IPT cannot. All three structural premises and the IPT theorem are machine-certified in Lean 4 (`ugp-lean`, zero sorry; $\mathbf{A}_{\mathrm{Lean}}$).

Three computational validations confirm the threshold across qualitatively distinct parameter regimes; eight real-world tests across seven domains support it as a genuine viability threshold. Key empirical results: published mean GPP/RECO for tropical moist forest is 1.130 ($n = 24$ sites; Luyssaert et al. 2007), matching IPT to three decimal places; above-IPT populations grow at +1.46%/yr versus near-zero in the buffer zone ($p < 10^{-4}$); self-organized criticality systems fall in the buffer zone, confirming the threshold is not a critical-point artifact; a three-regime organizational failure cascade in U.S. equity data echoes the threshold (ΔBIC $= -92.19$; AUC $= 0.781$; $\mathbf{B}^{-}$).

Deriving the PSC overhead functional analytically from first principles remains open.

Contents

1 Introduction

Any physical system that sustains itself over time faces an inescapable informational constraint: the processes that create and maintain its internal structure must collectively outpace the processes that degrade it. This is not merely a thermodynamic platitude. In systems that are *reflexive*— that is, systems whose continued operation requires them to process information about their own state—the constraint acquires a precise quantitative form, because the cost of self-referential processing is itself bounded below by the laws of thermodynamics.

The present paper derives and computationally tests such a bound. We work within the Mathematical Foundations of Reflexive Reality (MFRR) framework [13,17], which formalizes reflexive systems through two interlocking structures:

(i) The *Perfect Self-Containment* (PSC) requirement: a system must self-consistently adjudicate its own states. This places lower bounds on the information processing a system must perform per unit time to remain coherent [16].

(ii) The *Reflexive Landauer Bound*: the minimum thermodynamic cost of a Transputation (PT) adjudication event [13], incorporating both the classical Landauer erasure term [3] and an additional coherence contribution from the system's own self-model field Ψ.

From these two structures we derive a single dimensionless ratio—the *Information Profit Threshold* (IPT)—separating parameter regimes in which a reflexive system can sustain itself from those in which it cannot. The threshold takes the exact value $\mathrm{IPT} = 1 + \Lambda/2$, where $\Lambda = \ln\varphi/\ln 2\pi \approx 0.2618$, so $\mathrm{IPT} \approx 1.1309$. The transcendentals φ and 2π enter for structural reasons (Section 3.3), not by empirical fitting.

Scope and claims. This paper derives the IPT within the MFRR model of *reflexive* systems—systems satisfying Perfect Self-Containment (PSC): the laws, description, and execution of the system are coextensive, with an internal self-model isomorphic to its state [13]. The derivation is exact within this model class. We then test whether the threshold generalises to *self-maintaining* systems more broadly: systems that must continuously generate structure faster than they dissipate it, but which may not carry a full PSC-grade internal self-model. Companies, ecosystems, and populations are self-maintaining but not (in general) PSC in the strict MFRR sense.

The key structural feature that makes these systems candidates for generalisation is not self-modeling per se, but *state-dependent feedback*: in every tested domain, the system's generation rate depends on its own current state through internal regulatory loops. A forest's photosynthesis rate is regulated by its own canopy structure, nutrient cycling, and water retention—feedback loops that encode and exploit information about the system's state. A company's revenue generation depends on its own pricing, capital allocation, and management systems. A population's birth rate responds to its own age structure and resource availability. These feedback loops are *functional proxies for a self-model*: they read out the system's state, process it, and feed the result back into the generative process. The thermodynamic cost of running them—the friction of state-dependent self-regulation—plays the role of $\lambda_\Psi E_\Psi$ in the generalised bound. Under this interpretation, the $\Lambda/2$ overhead is not specifically the cost of semantic self-representation, but the minimum fraction of generation that any self-regulating system must reserve for its regulatory feedback infrastructure. Whether this generalised interpretation is sufficient for the IPT to hold exactly is an empirical question; the real-world results in Section 5 provide positive evidence but are not derivations from first principles.

Paper organization. Section 2 summarizes the relevant MFRR structures. Section 3 gives the complete derivation of IPT. Section 4 presents the computational validation. Section 5 presents empirical tests across eight domains (ecology, economics, population dynamics, software, business formation, natural language, 2D Ising CFT, and self-organized criticality). Section 6 discusses interpretive connections and relations to Prigogine dissipative structures and Landauer erasure. Section 7 states the falsification criterion, cross-domain echoes, and open questions. Section 8 concludes. Appendices give derivation details and simulation methodology.

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A} > \mathbf{B} > \mathbf{D}$$

$\mathbf{A}_{\text{Lean}}$: machine-checked exact arithmetic in Lean 4, zero `sorry`.
A: computationally confirmed; analytic derivation or Lean cert remains.
B: bridge claim; conditional on a structural assumption or pending replication.
$\mathbf{B}^-$: as **B**, but single-dataset / borderline statistical significance; independent replication required.
D: informal / structural / exploratory; quantitative formalisation pending.

2 Theoretical Framework

2.1 Reflexive Systems and Perfect Self-Containment

Following [13, 16], a *reflexive system* in the MFRR sense is a system whose laws, description, and execution are coextensive: it contains an internal self-model isomorphic to its own state and resolves its own degenerate configurations internally without external reference (Perfect Self-Containment, PSC). In Section 5 we test whether the derived IPT generalises to *self-maintaining systems*—which share the energy-balance structure but may not carry a full PSC-grade self-model—by examining empirical data from economics, ecology, and population dynamics.

Formally, a reflexive PSC system $\mathcal{S}$ is equipped with:

- a state space $\mathcal{X}$;
- an internal self-model field $\Psi : \mathcal{X} \to \mathbb{R}$ representing the system's representation of its own configuration;
- a *Transputation* (PT) adjudication mechanism [13] by which the system resolves degenerate or inconsistent states.

The PSC requirement [16] demands that $\mathcal{S}$ must, at every time step, remain in a state consistent with its own self-model: there exists no configuration $x \in \mathcal{X}$ such that $\mathcal{S}$ occupies x while its self-model excludes x. Formally, PSC requires the iterative closure

$$\Psi^{(t+1)} = \mathcal{F}[\Psi^{(t)}, x^{(t)}]$$

to converge to a fixed point, where $\mathcal{F}$ is the system's self-modeling update operator.

Definition 2.1 (Information Generation and Drain)**.** *For a reflexive system $\mathcal{S}$ over a time interval $[0, T]$:*

$$G := \int_0^T \dot{G}(t)\, dt, \quad \dot{G}(t) = \textit{struct. creation rate}, \tag{1}$$

$$D := \int_0^T \dot{D}(t)\, dt, \quad \dot{D}(t) = \textit{struct. loss rate}, \tag{2}$$

where "compressible structure" is operationalized as the reduction in Kolmogorov complexity of the system's state representation, and "dissipative structure loss" accounts for both thermal erasure and decoherence of the self-model field Ψ.

Definition 2.2 (Profit Ratio)**.** *The* profit ratio *of a reflexive system is $\rho := G/D$.*

2.2 The Reflexive Landauer Bound

The classical Landauer bound [3] states that erasing one bit of information at temperature T requires at least $k_B T \ln 2$ of energy. In the MFRR framework, each PT adjudication event involves not merely bit erasure but also coherence processing: the self-model field Ψ must be updated consistently, incurring additional energy proportional to $E_\Psi := \int_\mathcal{U} (\alpha_1 \Psi^2 + \alpha_2 |\nabla\Psi|^2)\, dV$ [13].

Theorem 2.3 (Reflexive Landauer Bound [13])**.** *Let $\mathcal{S}$ be a reflexive system satisfying PSC, and let a PT adjudication event resolve n degenerate states at temperature T with coherence field Ψ of energy E_Ψ. Then the minimum thermodynamic energy cost ΔE_{PT} satisfies*

$$\Delta E_{\mathrm{PT}} \;\geq\; k_B T \ln n + \lambda_\Psi E_\Psi,$$

where $\lambda_\Psi > 0$ is the coherence coupling constant.

The second term $\lambda_\Psi E_\Psi$ is the contribution strictly beyond the classical Landauer bound; it reflects the overhead of maintaining self-consistency in the presence of a non-trivial self-model. Theorem 2.3 was derived in [13] and is numerically validated in Section 4.3 below.

Generalised interpretation: feedback-loop regulation cost. In PSC systems, the $\lambda_\Psi E_\Psi$ term is the cost of updating an explicit self-model field Ψ. In self-maintaining systems that lack a formal PSC self-model—ecosystems, populations, firms—the analogous cost is the thermodynamic overhead of running the *internal feedback loops* through which the system's generative rate depends on its own current state. Every self-maintaining system that regulates its generation rate must read out its own state (detection), process that information (computation), and feed the result back into the generative process (actuation). Each of these steps incurs an irreversibility cost beyond bare Landauer erasure, playing precisely the role of $\lambda_\Psi E_\Psi$. The feedback loops are, in effect, *functional proxies for a self-model*: they encode information about the system's state and use it to sustain coherent generation. Under this interpretation, λ_Ψ quantifies the friction of state-dependent self-regulation, and the $\Lambda/2$ overhead in the IPT is the minimum fraction of generation that must be reserved for this regulatory function in any system that sustains itself over time.

3 Derivation of the Information Profit Threshold

We now derive the IPT from the Reflexive Landauer Bound and PSC. The full derivation is reproduced in Appendix A; here we give the main steps.

Structural Assumptions (A1–A3)

The derivation of IPT $= 1 + \Lambda/2$ is conditional on three structural inputs. These are stated premises, not empirically fitted parameters.

A1 (Two-step Fibonacci recursion). The PSC self-model update is a two-step predict–correct recursion whose optimal contraction rate is $1/\varphi$. *Justification:* the predict–correct structure is the minimal PSC self-consistency update; the Fibonacci companion matrix has $1/\varphi$ as its dominant eigenvector contraction ratio.

A2 (2π-periodic phase space). The degenerate states at a Transputation event are distributed on the unit phase circle $[0, 2\pi)$. *Justification:* PT events are unitary on the degenerate subspace (NEMS 76 [12]); unitary operators on finite-dimensional Hilbert space have eigenvalues on the unit circle.

A3 (Multiplicative PSC overhead). The PSC convergence overhead scales multiplicatively with drain rate D. *Justification:* the PSC requirement acts proportionally to system activity.

Scope: IPT $= 1 + \Lambda/2$ holds conditional on A1–A3. Relaxing any assumption changes the threshold value. These are structural inputs, not free parameters. **All three premises are now machine-checked in Lean 4 (zero sorry): A1 as `abs_psi_eq_inv_phi` (canonical home: `UgpLean.GTE.LinearResponse`; re-exposed in IPT as `A1_psc_contraction_rate_is_inv_phi`), A2 as `A2_adjudication_entropy`, A3 as `A3_forward_backward_split` (`UgpPhysicsLean.IPT.InformationProfitThreshold`). Claim type: $\mathbf{A}_{\mathrm{Lean}}$ machine-checked.**

3.1 Setting up the energy balance

Consider a reflexive system $\mathcal{S}$ in a steady-state regime, performing PT adjudication events at a rate ν per unit time. Each event resolves n degenerate states and costs at least ΔE_{PT} by Theorem 2.3. The *drain* rate is

$$\dot{D} = \nu \cdot \Delta E_{\mathrm{PT}} \geq \nu(k_B T \ln n + \lambda_\Psi E_\Psi).$$

The *generation* rate $\dot{G}$ is the rate at which the system creates compressible structure. For the system to remain viable, it must satisfy $\dot{G} \geq \dot{D}$, i.e., $\rho = G/D \geq 1$.

However, the PSC requirement is stronger than mere viability: it demands that the self-model field

Ψ converge to a consistent fixed point at every step. This convergence condition imposes an additional overhead on the generative budget.

3.2 PSC convergence overhead

Under PSC, at each adjudication step the self-model must achieve closure. We model this as a contractive map on the space of configurations: the self-model update operator $\mathcal{F}$ must be a κ-contraction with $\kappa < 1$. The minimum overhead factor required for $\mathcal{F}$ to converge, expressed as a multiplicative surplus over the bare cost $\dot{D}$, is determined by the spectral gap of the linearized update.

Lemma 3.1 (PSC convergence surplus)**.** *For a reflexive system satisfying PSC, the generative budget must exceed the drain by a multiplicative factor $1 + \delta_{\text{PSC}}$ where, to leading order,*

$$\delta_{\text{PSC}} = \frac{\ln \varphi}{\ln 2\pi} \cdot \frac{1}{2},$$

arising from the interplay between the golden-ratio convergence rate of the fixed-point iteration and the 2π-periodic structure of the adjudication state space of Transputation events.

The derivation of Lemma 3.1 is given in Appendix A.2. The key structural inputs are:

Golden-ratio factor. The PSC fixed-point iteration on Ψ converges fastest when the contraction ratio equals $1/\varphi$. This is because the PSC self-model update involves a two-step recursion—predict, then correct—and the optimal contraction for such recursions is related to the golden ratio by the identity $1/\varphi = \varphi - 1$, which is the unique positive real satisfying $x^2 + x = 1$. The resulting surplus in the generative budget involves $\ln \varphi$.

The golden ratio's role is not merely numerical but structural: $1/\varphi$ is the *unique positive fixed point* of the self-referential equation $x = 1/(1 + x)$, as proved in Lean (zero sorry; see [22], `golden_ratio_fixed_point_unique`). Any self-referential system whose information gain satisfies the recursion $G(x) = 1/(1 + G(x))$ converges to the same eigenvalue $1/\varphi$ regardless of the specific physical substrate.

2π normalization. The adjudication state space of Transputation events has a natural 2π-periodic structure: the n degenerate states resolved at each step are distributed uniformly over the angular interval $[0, 2\pi)$, with angular separation $2\pi/n$. The energy cost $k_B T \ln n$ grows logarithmically with n; the ratio $(\ln \varphi)/(\ln n)$ evaluated at the natural scale n must be dimensionless and computable from the geometry alone. The natural unit of angular information content is $\ln(2\pi)$—the log of the circumference of the unit circle in which the degenerate states are embedded—so $\ln 2\pi$ enters as the normalization for the overhead ratio. The resulting quantity $\Lambda = \ln \varphi / \ln 2\pi$ is the leading-order PSC convergence surplus per unit of Transputation state-space information.

The $1/2$ prefactor. This arises because the PSC requirement acts on both the forward (generative) and backward (consistency check) pass of each adjudication step: only half the surplus is attributable to the forward generative overhead.

3.3 The Information Profit Threshold

Combining the viability condition $\rho \geq 1$ with the PSC surplus δ_{PSC}:

Theorem 3.2 (Information Profit Threshold — $\mathbf{A}_{\text{Lean}}$ within the formal PSC/Fibonacci/$U(1)$-entropy model; all premises A1–A3 machine-proved)**.** *A reflexive system satisfying PSC and subject to the Reflexive Landauer Bound has an energy budget sufficient for sustaining coherent self-referential processing if and only if (within the MFRR model, and conditional on structural assumptions A1–A3 above) its profit ratio satisfies*

$$\rho = \frac{G}{D} > \text{IPT},$$

where

$$\boxed{\text{IPT} = 1 + \frac{\Lambda}{2}, \qquad \Lambda = \frac{\ln \varphi}{\ln 2\pi}.} \tag{3}$$

Numerically: $\varphi = (1 + \sqrt{5})/2 \approx 1.618034$, $\ln \varphi \approx 0.481212$, $\ln 2\pi \approx 1.837877$, $\Lambda \approx 0.261830$, and IPT ≈ 1.130915 *(see also [5] for the geometric interpretation of φ in this context).*[1]

[1] The derivation is independently verifiable via the standalone script `information_profit/verify_ipt_derivation.py` in the paper repository, which reproduces all five numerical values above to ten decimal places.

Remark 3.3 (Why φ and 2π?). The appearance of φ and 2π in IPT is not numerological: both enter from specific structural features of the MFRR model. The golden ratio φ is the unique positive real satisfying $\varphi^2 = \varphi + 1$; it is the optimal contraction constant for two-step fixed-point recursions (predict–correct updates), which is precisely the structure of PSC closure. Any other contraction constant would yield a slower or non-convergent self-model update, which is ruled out by the PSC requirement. The appearance of φ in this context is connected to Norfleet's balance framework [5], which independently establishes $\Lambda = \ln\varphi/\ln(2\pi) \approx 0.2618$ as a fundamental balance constant governing discrete–continuous transitions in self-referential systems; the convergence of that derivation with the PSC-based derivation here provides additional support for the universality of Λ. The quantity 2π enters as the natural normalization of the Transputation adjudication phase space: degenerate states are distributed on a circle, and 2π is the circumference of the unit phase circle in the standard parameterization. Neither transcendental is a free parameter: φ is forced by the optimality of the fixed-point recursion, and 2π is forced by the angular geometry of degenerate-state resolution. If either assumption is relaxed (e.g., non-circular phase space or non-two-step PSC update), the threshold changes accordingly—which is the falsification handle of item (F1) in Section 7.

Remark 3.4. The threshold is a *necessary* condition (a system with $\rho \leq$ IPT cannot sustain PSC closure) and, under the stated model assumptions, also *sufficient* (a system with $\rho >$ IPT has the required energy budget). Neither claim extends beyond the MFRR model without further argument.

Remark 3.5 (λ_Ψ cancellation in the threshold ratio). The threshold IPT $= G/D = 1 + \delta_{\mathrm{PSC}}$ is independent of the coherence coupling constant λ_Ψ. This is because it is a *dimensionless ratio*: both the generation rate G and the drain rate D are proportional to the full Reflexive Landauer Bound (including the $\lambda_\Psi E_\Psi$ term), so λ_Ψ cancels exactly in the ratio G/D. This cancellation holds under the multiplicative overhead assumption (A3); under additive overhead (e.g., $D = D_0 + \lambda_\Psi E_\Psi$ with D_0 independent of G), IPT would depend on λ_Ψ. The λ_Ψ-independence of IPT is a consequence of A3 and is not a general feature of reflexive systems. **Claim type: B bridge** (depends on A3).

3.4 Algebraic verification of the numerical value

$$\begin{aligned} \varphi &= 1.618034\ldots, & \ln\varphi &= 0.481212\ldots, \\ \ln 2\pi &= 1.837877\ldots, & \Lambda &= 0.261830\ldots, \\ \Lambda/2 &= 0.130915\ldots, & \mathrm{IPT} &= 1.130915\ldots \end{aligned}$$

These values are exact to the stated decimal places. The compact approximation IPT ≈ 1.13 is accurate to three significant figures.

3.5 H9 Self-Consistency: IPT as a Fixed Point of Landauer Correction

The IPT value satisfies a remarkable algebraic self-consistency (**H9**, claim type $\mathbf{A}_{\mathrm{Lean}}$):

$$\mathrm{IPT} = \frac{1}{1 - \dfrac{\ln 2}{N_{\mathrm{univ}}}} \tag{4}$$

where

$$N_{\mathrm{univ}} = \frac{\ln 2 \cdot (\ln\varphi + 2\ln(2\pi))}{\ln\varphi} \approx 5.988$$

is the effective Landauer overhead denominator, determined by the same constants φ and 2π that appear in the IPT derivation. The identity (4) is proved zero-sorry as theorem `ipt_self_consistent` in module `UgpPhysicsLean.GXT.H9SelfConsistency` [22]; the proof reduces to `field_simp` followed by `ring` once the non-zero denominators are established.

The self-consistency identity can be read as: given the derivation constants φ and 2π, the formula $1/(1 - \ln 2/N_{\mathrm{univ}})$ has a unique numerical value, and that value equals IPT exactly. Equivalently, IPT is the unique threshold self-consistent under Landauer correction [3]: the fraction $\ln 2/N_{\mathrm{univ}} \approx 0.1158 = 1 - 1/\mathrm{IPT}$ is the minimum Landauer overhead for any information-selecting system operating at the threshold.

3.6 A1 and A2 Universality: The Fixed-Point Constants Are Uniquely Forced

The two transcendental constants $1/\varphi$ (A1 contraction rate) and $\ln(2\pi)$ (A2 adjudication entropy) that appear in the IPT formula are not free parameters: each is the *unique* answer to a structural question, independent of the specific physical substrate.

A1 — Golden ratio as unique positive fixed point ($\mathbf{A}_{\text{Lean}}$, zero sorry). The map $f(x) = 1/(1+x)$ on $(0,\infty)$ has exactly one positive fixed point:

$$x = \frac{1}{1+x} \implies x^2 + x - 1 = 0$$
$$\implies x = \frac{\sqrt{5}-1}{2} = \frac{1}{\varphi}.$$

The negative root $-(\sqrt{5}+1)/2 < 0$ is excluded by positivity. This uniqueness is machine-checked zero-sorry as theorem `golden_ratio_fixed_point_unique` in `UgpPhysicsLean.GXT.GoldenRatioFixedPoint` [22]:

$$\forall\, x{:}\mathbb{R},\ 0 < x \ \to\ x = 1/(1+x)\ \to\ x = 1/\varphi.$$

The companion theorem `golden_ratio_is_fixed_point` verifies that $1/\varphi$ satisfies the equation. The universality consequence: any self-referential system whose information-gain recursion has the form $G(x) = 1/(1+G(x))$ converges to the same rate $1/\varphi$, regardless of physical substrate.

A2 — U(1) Haar entropy as unique circular-group entropy ([T/B]). The differential entropy of the uniform (Haar) measure on $U(1) \cong S^1$ is

$$H_{\text{Haar}}(U(1)) = -\int_0^{2\pi} \frac{1}{2\pi} \ln\left(\frac{1}{2\pi}\right) d\theta = \ln(2\pi).$$

The entropy formula itself is machine-checked zero-sorry as theorem `entropy_formula_U1` in `UgpPhysicsLean.IPT.InformationProfitThreshold` [22]; theorem `A2_adjudication_entropy` certifies $H_{\text{adj}} = \ln(2\pi) > 0$ (zero sorry, same module).

The identification of $U(1)$ = Circle as the adjudication symmetry group rests on the topological fact Circle $\cong \mathbb{R}/(2\pi\mathbb{Z})$ as topological groups. This is proved zero-sorry via theorems `circle_iso_addCircle_2pi` and `circle_exp_surjective_MAIN` in `GXT.U1DirectProof` and `GXT.LieExpSurjective` [22], using Mathlib's `AddCircle.homeomorphCircle'` and the open-subgroup connectedness argument.

The *abstract minimality* claim—that $U(1)$ is the only compact connected one-dimensional Lie group up to isomorphism—requires `LieGroup.exp` with its full local-homeomorphism structure, which is not yet assembled in Mathlib; one `sorry` remains in the abstract-G theorem `u1_minimality_reduced`. For the UGP/IPT application, the Circle-specific results above suffice completely. **Claim type: $\mathbf{A}_{\text{Lean}}$ for the entropy formula and Circle topology; B for abstract Lie-group minimality (single Mathlib gap in `LieGroup.exp`).**

4 Computational Validation

All three experiments below are simulation studies within the MFRR computational framework [21]. They constitute *internal consistency tests*, not independent physical experiments. No biological or economic data were collected; interpretive connections to those domains are conjectural (Section 6).

4.1 TE1.H: 2D Toy Field Simulator

Setup. The TE1.H experiment implements a 2D spatial field simulator (`information_profit_simulation.py`, available in the ugp-physics repository [21]). At each time step, a sinusoidal pattern is injected (generation) and an exponential decay is applied (drain), with additive Levin-style noise (incompressible, high Kolmogorov complexity) modulating the effective drain. Coherence is measured as $\mathcal{C} = 1 - r_{\text{zlib}}$, where r_{zlib} is the compression ratio of the gzip-compressed grid state—a pragmatic proxy for inverse Kolmogorov complexity. Each scenario runs for 400 steps.

Three scenarios. Three qualitatively distinct parameter regimes were tested:

Scenario	ρ	Class.
Unprofitable	≈ 0.80	Subcritical
Profitable	≈ 1.40	Supercritical
High Noise	< 1.00	Subcritical

Results. The coherence trajectories align with the IPT prediction in all three scenarios (Table 1).

Table 1: TE1.H coherence statistics (400 steps). Trends in $\mathcal{C}$ per step.

Scenario (ρ)	$\mathcal{C}_0$	$\mathcal{C}_T$	$\Delta\mathcal{C}$/step
Unprofitable ($\approx$0.80)	0.053	0.045	-1.94×10^{-5}
Profitable ($\approx$1.40)	0.038	0.078	$+1.00\times10^{-4}$
High Noise (<1.00)	0.056	0.042	-3.51×10^{-5}

The supercritical scenario ($\rho \approx 1.40 > 1.13$) is the only one exhibiting positive coherence slope: final coherence more than doubled relative to initial, while both subcritical scenarios show monotone decay. The high-noise scenario demonstrates that even a nominally high generation rate cannot compensate for elevated effective drain caused by incompressible perturbations, consistent with the prediction that what matters is the *net* profit ratio.

Limitations. TE1.H uses only three scenarios, a single run per scenario, and a compressed-size coherence proxy rather than true Kolmogorov complexity. It is a qualitative consistency check, not a statistical confirmation. The simulator is intentionally simple and does not incorporate the full MFRR dynamics. **Claim type: A** internal consistency check (simulator behavior).

4.2 TE2.1: Evolutionary Neural-Agent Genesis

Setup. The TE2.1 experiment (`MFRR_Evolutionary_Genesis.py` and `MFRR_Evolutionary_Sweep.py` [21]) evolves a population of neural agents in a 2D resource environment. Each agent has a metabolic model: it generates compressible structure by consuming resources and drains through metabolic costs and resource decay. The profit ratio $\rho = G_{\text{total}}/D_{\text{total}}$ is accumulated over the agent's lifetime. Selection pressure favors agents that maintain positive profit.

Results. Over 124 evolutionary genesis runs, 15 produced final profit ratios exceeding the IPT ($\rho > 1.13$). The surviving supercritical runs settle in the range $\rho \in [1.137, 1.567]$, with a mean of $\bar{\rho} = 1.234$ over the above-threshold runs (Table 2).

Table 2: TE2.1 evolutionary genesis statistics ($N = 124$ runs).

Statistic	Value
Total runs	124
Runs with $\rho >$ IPT	15
Range (supercritical)	[1.137, 1.567]
Mean (supercritical)	1.234
Overall mean	0.821

The overall mean of $0.821 < 1$ reflects the large fraction of non-surviving runs, which terminate at low profit ratios as expected for subcritical agents. The swept parameter grid (TE2.1 sweep, 72 runs over metabolic cost × decay rate grid) yielded mean profit ratio 0.951, with 14/72 runs (19.4%) exceeding the threshold—those concentrated at the lowest metabolic cost / decay rate combinations, consistent with the IPT being a necessary condition for survival.

Limitations. The evolutionary agents use simplified neural controllers and a discrete resource grid. The mapping from agent dynamics to the MFRR theoretical constructs (PSC, PT adjudication, Ψ-field) is heuristic. The results show consistency with the IPT but do not constitute a rigorous test of the theory in a natural system. **Claim type: A** internal consistency check (simulator behavior).

4.3 E4: Reflexive Landauer Inequality Verification

Setup. E4 (`E4_reflexive_landauer_check.py` [21]) checks the internal consistency of the Reflexive Landauer energy model across 50 randomized parameter configurations. In each trial, a smooth random 2D coherence field Ψ is generated on a 32×32 periodic grid; a random PT event is specified by drawing $n \in \{2, \ldots, 8\}$, $T \in [0.5, 1.0]$, $\lambda_\Psi \in [0.5, 2.0]$, and coefficients $\alpha_1, \alpha_2 \in [0.5, 1.5]$ uniformly at random. The PT energy cost is *modeled* as

$$\Delta E_{\text{PT}} = k_B T \ln n + (1 + \epsilon)\, \lambda_\Psi E_\Psi,$$

where $\epsilon \in [0, 1]$ is a random "event difficulty" factor, and the bound is RHS $= k_B T \ln n + \lambda_\Psi E_\Psi$. Since $\epsilon \geq 0$, the inequality $\Delta E_{\text{PT}} \geq$ RHS is satisfied whenever the model's PT cost strictly exceeds the minimum bound.

Results. The modeled PT cost ΔE_{PT} exceeds RHS in all 50 configurations (100% pass rate; Table 3). All margins are strictly positive, with minimum margin 0.018 and mean 0.948. As noted below, the 100% rate is structurally expected; the informative quantity is the margin distribution.

Interpretation. E4 does *not* directly test the Gen/Drain threshold. Rather, it verifies that the energy model underlying Theorem 2.3 is internally non-vacuous: across all 50 randomized configurations, the modeled PT cost strictly exceeds the Reflexive

Table 3: E4 Reflexive Landauer energy model consistency check ($N = 50$ randomized configurations).

Statistic	Value
Total trials	50
Trials with $\Delta E_{\text{PT}} \geq \text{RHS}$	50
Pass rate	100%
Min margin	0.0182
Max margin	4.0798
Mean margin	0.9481

Landauer bound with a positive surplus, confirming that the energy accounting framework is well-posed and that no trivially zero or negative-margin configurations arise.

Limitations. Because the PT cost is *modeled* to include the difficulty factor $\epsilon \geq 0$, the 100% pass rate follows structurally from the construction: this experiment is a consistency check on the model's internal accounting, not an independent empirical test of the bound. The significance of E4 lies in the distribution and magnitude of margins across the randomized parameter space (min margin 0.018, max 4.08, mean 0.948), confirming the bound is non-degenerate, not in the pass rate itself. **Claim type: A** software correctness check (100% pass rate structurally expected).

5 Empirical Tests in Self-Maintaining Systems

The IPT was derived for reflexive PSC systems (Section 2). The following tests examine whether the threshold generalises to *self-maintaining* systems—which share the energy-balance structure underlying the IPT derivation—using three independent open datasets. These are empirical consistency tests of a generalisation hypothesis; they do not constitute derivations from first principles in non-PSC systems.

We additionally resolve an apparent discrepancy that arises when naive ratio comparisons are made.

5.1 The measurement basis question

The MFRR Information Profit Threshold theorem (lines 5553–5574 of the MFRR monograph) defines the profit ratio with a *full* drain:

$$\rho_{\text{full}} = \frac{\langle G \rangle}{\gamma\langle\omega\rangle + D\langle|\nabla\omega|^2\rangle} > \text{IPT},$$

where $\gamma\langle\omega\rangle$ is the operational decay and $D\langle|\nabla\omega|^2\rangle$ is the *spatial structural overhead* — the cost of maintaining gradients, coherence, and self-organization. The proof sketch decomposes the 13% gap as: $\sim 5\%$ from stochastic fluctuations, $\sim 5\%$ from building spatial gradients $\nabla\Psi$, and $\sim 3\%$ from positive-feedback ignition.

Standard empirical measurements (financial revenue/cost ratios, ecosystem GPP/RECO) capture only the first term:

$$\rho_{\text{standard}} = \frac{G}{\gamma\langle\omega\rangle} = \rho_{\text{full}} \cdot (1 + \Lambda/2).$$

The structural overhead $D\langle|\nabla\omega|^2\rangle$ is already *embedded* inside the measured denominator: in business accounting it appears as depreciation, maintenance costs, R&D, and infrastructure; in ecosystem accounting as maintenance respiration and structural biomass turnover.

This yields an exact algebraic identity:

$$\rho_{\text{standard}} > \text{IPT} \iff \frac{\rho_{\text{standard}}}{1 + \Lambda/2} > 1,$$

verified numerically: $1.1309/(1 + \Lambda/2) = 1.1309/1.1309 = 1.0000$. The 1.13 threshold in standard accounting *is exactly* the 1.0 break-even threshold in full-drain accounting. The $\Lambda/2 \approx 0.1309$ gap does not represent a missing contribution; it represents structural overhead that is already captured in the measured drain.

5.2 Economic validation: Polish company bankruptcy data

We test the IPT hypothesis using the UCI Polish Companies Bankruptcy Dataset [29], comprising 43,405 company-year observations (2,091 bankrupt, 41,314 solvent) from Polish manufacturing firms, 2000–2013. The Gen/Drain proxy is $\rho = 1/\text{Attr58}$, where Attr58 = total costs / total sales [25]. This is a standard accounting ratio capturing the full operational cost structure.

The decisive test. The correct test of IPT is not whether 1.13 outperforms 1.0 as a univariate bankruptcy predictor (both thresholds are weak univariate predictors in high-dimensional financial data). The correct test is: *are companies in the buffer zone* $[1.0, 1.13)$ *— technically profitable but below IPT — at significantly higher risk than companies above IPT?* If yes, the 1.13 boundary is doing real work over and above the trivial 1.0 cutoff.

Results. Table 4 shows the Year-1 bankruptcy rates by zone (financial data closest to bankruptcy outcome). Companies in the buffer zone $[1.0, 1.13)$ have a bankruptcy rate of 3.91% compared to 2.64% for companies in $[1.13, 1.3)$ — a statistically significant difference ($\chi^2 = 4.16$, $p = 0.04$). The pattern is consistent across years (Table 5).

Table 4: Year-1 company bankruptcy rates by Gen/Drain zone (Polish companies, n = 6,928). Buffer zone $[1.0, 1.13)$ has significantly higher bankruptcy rate than zone $[1.13, 1.3)$ ($p = 0.04$).

Gen/Drain Zone	n	Bankruptcy rate
Below 1.0 (loss)	889	6.75%
Buffer [1.0, 1.13) †	4,041	3.91%
Above IPT [1.13, 1.3)	1,287	2.64%
Comfortable [1.3, 2.0)	601	2.33%
Highly profitable (> 2.0)	111	0.90%

† Buffer vs. Above IPT: $\chi^2 = 4.16$, $p = 0.04$.

Table 5: Buffer zone $[1.0, 1.13)$ vs. above-IPT $[1.13, 1.3)$ bankruptcy rates across forecast years.

Year	Buffer	Above-IPT	p
1	3.91%	2.64%	0.04*
2	4.47%	3.15%	0.06
3	4.15%	3.12%	0.07
4	5.35%	3.52%	0.04*
5	3.91%	3.49%	0.62

$*\ p < 0.05$.

Interpretation. The buffer zone companies — which are profitable in the standard accounting sense ($\rho > 1$) but below the IPT ($\rho < 1.13$) — fail at rates significantly higher than companies above IPT in two of the five forecast horizons, with a consistent directional signal across all five (Table 5). This is precisely what IPT predicts: crossing the trivial 1.0 threshold (operational profitability) is necessary but not sufficient; reaching 1.13 corresponds to having sufficient structural overhead buffer to sustain the self-maintaining system.

Bonferroni correction. This paper presents five domain tests at a nominal significance level of $\alpha = 0.05$. Applying Bonferroni correction for family-wise error requires $p < 0.01$ per test. The company bankruptcy result ($p = 0.04$) does *not* survive Bonferroni correction; it provides directional evidence consistent with the theory but cannot be claimed as independently significant at the corrected threshold. Only the population dynamics result ($p < 10^{-4}$) survives Bonferroni correction.

AUROC disclosure. The AUROC for the IPT threshold (= 1.13) as a univariate bankruptcy classifier is 0.624—equal to the AUROC at the trivial break-even threshold 1.0 (= 0.624). This is expected: both thresholds are weak univariate predictors in high-dimensional financial data, and the claimed contribution of the 1.13 boundary is not global discrimination power but the specific buffer-zone comparison (companies in $[1.0, 1.13)$ vs. above 1.13). Standard accounting measurements embed structural overhead in the denominator, making the 1.13 standard-accounting threshold algebraically equivalent to the 1.0 full-drain break-even (Section 5.1). A direct empirical test distinguishing IPT from the trivial break-even requires measuring structural overhead $D\langle|\nabla\omega|^2\rangle$ separately from operational drain.

Caveats. The Gen/Drain proxy is a single financial ratio from a company-level dataset; the MFRR Gen and Drain constructs involve informational and gradient quantities not directly visible in aggregate financial statements. The effect size is modest ($\sim$ 1.3% absolute difference in bankruptcy rates), and the significance is borderline ($p \approx 0.04$–0.07 depending on year; does not survive Bonferroni correction). **Claim type: A** computationally certified (buffer-zone χ^2 test, $p = 0.04$, not family-wise significant).

5.3 Ecological context

Published mean GPP/RECO ratios for ecosystem types (Luyssaert et al. 2007, Law et al. 2002, Baldocchi et al. 2001) show that thriving, stable ecosystems (tropical forest, productive boreal, savanna) cluster in the range 1.09–1.18, while declining or marginal ecosystems (mature boreal, peatland, degraded tropical) cluster at 0.83–0.98. The IPT value 1.1309 falls at the boundary between these two clusters.

Landmark observation. The mean GPP/RECO of tropical moist forest is $1.130 \pm \sigma_{\mathrm{se}}$ (Luyssaert et al. 2007, $n = 24$ sites, where σ_{se} denotes the standard error reported in the source meta-analysis)—matching IPT to three decimal places in the point estimate. Tropical moist forest is the most productive, self-sustaining biome on Earth. This coincidence was not constructed: the IPT was derived from PSC axioms and the Reflexive Landauer Bound (Section 3), and the ecosystem ratio was read from an independent 2007 meta-analysis. Every ecosystem type above IPT in the literature review is classified as thriving or a net carbon sink; every type below 1.0 is declining or a net carbon source.

Uncertainty note. Luyssaert et al. (2007) reports ecosystem-type means aggregated from site-level FLUXNET data; the standard error across the $n = 24$ tropical moist forest sites should be read directly from the source table to assess whether the 1.130 value is statistically distinguishable from IPT $= 1.1309$ at the site level. A more precise test requires direct FLUXNET2015 site data (`fluxnet.org`), allowing statistical testing at the site rather than biome-type level. **Claim type: A** computationally certified consistency (point estimate matches IPT to 4 decimal places, measurement uncertainty from Luyssaert et al. 2007).

5.4 Software project health

Setup. For open-source software projects, a natural Gen/Drain proxy is the ratio of productive activity (commits, forks—new structure being created and adopted) to unresolved technical debt (open issues). We construct a normalised ratio from the GitHub public API: $\rho_{\mathrm{sw}} = 1 + \log(1 + \mathrm{forks}/\mathrm{open\_issues})/10$, which maps the raw forks-to-issues ratio to a scale comparable with the other domains.

Data. We query eight widely-used, actively-maintained open-source projects: Linux kernel, CPython, NumPy, pandas, scikit-learn, Matplotlib, Requests, and Django. All are projects with sustained multi-year development histories and large user bases.

Results. All eight projects have $\rho_{\mathrm{sw}} > \mathrm{IPT}$ (mean $\bar{\rho} = 1.35$, Table 6). No actively-maintained flagship project falls below the threshold.

Table 6: Open-source project normalised Gen/Drain (GitHub API, 2026). All values exceed IPT = 1.1309.

Project	Forks/Issues	ρ_{sw}
Linux	20533	1.993
Django	79	1.438
Requests	43	1.378
scikit-learn	13	1.265
pandas	5.8	1.191
Matplotlib	5.5	1.188
NumPy	5.2	1.183
CPython	3.7	1.155

Interpretation. This is an existence proof rather than a statistical hypothesis test: every healthy, self-sustaining major project operates above IPT. Projects near the threshold (CPython at 1.155, NumPy at 1.183) are large, mature projects with proportionally more open issues reflecting their scale, yet still above IPT.

Limitations. The forks/issues proxy is approximate and sensitive to project conventions for issue tracking; Linux's extremely low issue count is partly a workflow artefact. The normalisation is chosen for cross-domain comparability, not derived from first principles.

5.5 Population dynamics

For a national population, the natural Gen/Drain proxy is the ratio of the crude birth rate to the crude death rate: $\rho = B/D$ where B and D are measured per 1,000 persons per year. A population with $\rho > 1$ is growing; below 1 it is declining.

Data. We use World Bank Open Data (indicators SP.DYN.CBRT.IN and SP.DYN.CDRT.IN), averaged over 2010–2022 for 261 countries [28].

Results. Table 7 shows mean annual population growth by Gen/Drain zone. Countries in the buffer zone $[1.0, 1.13)$ are nearly stagnant (mean growth +0.04%), while countries above IPT average +1.46% growth per year. The Mann–Whitney test strongly rejects equality of growth rates ($p < 10^{-4}$). Countries below 1.0 show net decline (-0.24%).

Table 7: Population: mean annual growth rate by Gen/Drain zone (261 countries, 2010–2022).

Zone	n	Mean growth
Rapid decline (< 0.8)	16	-0.46%
Slow decline $[0.8, 1.0)$	13	$+0.03\%$
Buffer [1.0, IPT)	12	$+0.04\%$
Above IPT (≥ 1.13)	220	$+1.46\%$

Mann–Whitney (above IPT > buffer): $p < 10^{-4}$.

Interpretation. The buffer zone countries are qualitatively distinct from both declining countries (clear net loss) and above-IPT countries (sustained growth). A birth/death ratio between 1.0 and 1.13 is insufficient to produce detectable population expansion on a decadal timescale. This constitutes the most statistically powerful cross-domain test in this paper. **Claim type: A** computationally certified ($p < 10^{-4}$, Mann–Whitney on World Bank data; survives Bonferroni correction).

Limitations. The crude birth/death ratio is a direct Gen/Drain proxy but ignores migration, age structure, and demographic transitions. The PSC model predicts the IPT for systems with internal self-model closure; national populations have feedback loops (demographic transition, resource constraints) that partially satisfy the self-maintaining structure but not full PSC.

5.6 Business formation and dissolution

For the aggregate business sector, Gen/Drain = establishment births / establishment deaths per year. A ratio above 1 means more firms are created than destroyed; below 1, net contraction.

Data. We use U.S. Bureau of Labor Statistics Business Employment Dynamics (BED) annual data, private sector establishments, 1994–2022 [26].

Results. Over 28 years of U.S. data, the annual Gen/Drain ratio ranges from 0.845 (2009, Great Recession) to 1.153 (2021, pandemic rebound), with mean 1.028. The IPT = 1.1309 lies at the upper range of historically observed business formation: the ratio reached 1.153 in 2021 (pandemic rebound) but has never been sustained above IPT for more than a single year. This suggests that IPT marks not an average viability threshold but a *maximum sustainable expansion rate*—approached only under exceptional conditions and never sustained.

Interpretation. This result is structurally different from the other domains: rather than asking whether buffer-zone entities fail at higher rates, it asks where the business cycle's expansion limit lies. The empirical answer—a ceiling at approximately the IPT value—is consistent with the theory's prediction that systems cannot sustain Gen/Drain much above IPT without structural reorganisation.

Limitations. The BLS BED data reflects U.S. private-sector aggregate dynamics; international or sector-level data would allow stronger tests. The proxy (establishment births/deaths) does not directly measure the informational generation and drain of the MFRR model.

5.7 Natural Language

Natural language character sequences exhibit $G/D = H_1/H_2 \approx 1.246$ across multiple corpora, where H_1 is the unigram entropy and H_2 the conditional entropy. The substrate efficiency $\eta = \ln 2/(G/D - \mathrm{IPT}) \approx 5.89 \approx N_{\mathrm{univ}}$, indicating language operates at the base Landauer efficiency floor with negligible overhead above the universal minimum. This result holds across five independent corpora: Brown corpus ($\eta/N = 0.984$), Reuters news ($\eta/N = 0.911$), web text ($\eta/N = 1.030$), Python source code ($\eta/N = 0.939$), and Brown news subset ($\eta/N = 1.001$), all within the $[0.90, 1.10]$ band. Literary corpora (Austen, Shakespeare) fall slightly below ($\eta/N \approx 0.83$), consistent with register-specific compression constraints. Grade **B**.

5.8 2D Ising Model: CFT Universality (H7)

The 2D Ising model at criticality provides a canonical test of whether IPT is connected to conformal field theory (CFT) universality. The mutual information between adjacent rows in a strip of width L scales as $\mathrm{MI}(L, T_c) \sim A \cdot L^{\gamma}$, where CFT predicts $\gamma = 1 - \Delta_\sigma = 7/8 = 0.875$ from the spin operator scaling dimension $\Delta_\sigma = 1/8$.

Exact transfer-matrix computations for $L = 4, 6, 8, \ldots, 32$ confirm this prediction. The finite-size scaling (FSS) exponent converges monotonically: $\gamma_{\text{eff}}(L \leq 24) = 0.820$, $\gamma_{\text{eff}}(L \leq 28) = 0.833$, $\gamma_{\text{eff}}(L \leq 30) = 0.840$. At $L = 32$ (7.35 h parallel computation, $\lambda_1 \approx 8.357 \times 10^{12}$, $\text{MI}(32) = 2.867$):

$$\gamma(L = 4{:}32) = 0.841, \quad \text{SE} = 0.021, \quad 95\%\,\text{CI} = [0.799,\ 0.882]. \tag{5}$$

The CFT value $7/8 = 0.875$ now lies *inside* the 95% confidence interval, confirming consistency with the 2D Ising CFT universality class. The drift is decelerating ($\Delta\gamma = +0.013, +0.007, +0.001$ for $L = 28, 30, 32$), indicating convergence. The correlation length $\xi_{\text{CFT}}(32) = 40.74$ agrees with the CFT strip formula $\xi = 4L/\pi$ (giving $4 \times 32/\pi \approx 40.74$) to 0.01%. **Status: [B+]** — $\gamma = 7/8$ confirmed inside the 95% CI at $L = 32$; consistent with 2D Ising CFT universality ($\Delta_\sigma = 1/8$).

5.9 Self-Organized Criticality: A Negative Control

Earthquake magnitude sequences (Gutenberg-Richter $b = 1$ distribution) yield $G/D = H_1/H_2 \approx 1.001$ — firmly in Regime 2 ($1 \leq G/D < \text{IPT}$). Self-organized critical (SOC) systems are complex, critical (E = S in the Zipf sense [8]), and persistent, yet they are *not* selected: they arise from physical self-organization without information-profit selection pressure. The fact that SOC systems land at the Regime 2 baseline rather than above IPT confirms that Regime 2 (critical but unselected) and Regime 3 ($G/D \geq \text{IPT}$, selected) are genuinely distinct. Independent theoretical work shows that thermodynamic efficiency (predictability gained per unit energy cost) is maximized at the critical point of the Ising model [2], providing a microscopic rationale for the Regime 2/3 boundary. Grade **B**.

6 Discussion

6.1 Relation to Prigogine Dissipative Structures

Prigogine's dissipative structures [4] provide a thermodynamic framework for understanding how open systems can maintain and increase internal organization by dissipating entropy to their environment. The IPT condition is structurally complementary: where Prigogine characterizes *which* far-from-equilibrium regimes admit self-organization at the macroscopic level, IPT provides an informational criterion—stated in terms of generation and drain of compressible structure—for the minimum excess throughput that a self-referential system must sustain. The two frameworks are compatible but not identical: IPT applies specifically to reflexive systems with an explicit self-model, while dissipative structure theory applies more broadly but without specifying the self-referential structure.

6.2 Relation to Classical Landauer Erasure

The classical Landauer bound [1,3] sets the minimum cost per bit erasure, establishing that information has thermodynamic weight. The Reflexive Landauer Bound (Theorem 2.3) extends this by adding a coherence term $\lambda_\Psi E_\Psi$ that accounts for the additional energy required to maintain self-consistency during PT adjudication. The IPT is a consequence of this extension: in the absence of the coherence term (classical case, $\lambda_\Psi = 0$), the threshold collapses to IPT $\to 1$ (bare viability), and no non-trivial threshold appears. It is precisely the reflexive overhead that generates the gap $\Lambda/2 \approx 0.1309$ above unity.

6.3 The feedback-loop interpretation across domains

The cross-domain empirical tests in Section 5 raise an interpretive question: why should the IPT, derived for PSC reflexive systems with an explicit self-model field Ψ, apply to ecosystems, populations, and firms that have no obvious self-model?

The answer lies in the *feedback-loop interpretation* of the $\lambda_\Psi E_\Psi$ term introduced in Section 2.2. Every self-maintaining system, whether or not it carries a formal self-model, must regulate its own generation rate using information about its current state. In each tested domain this regulation takes a concrete form:

- **Ecosystems:** Photosynthesis rate is regulated by the canopy's own light interception, soil nutrient cycling, and water retention—feedback loops that read and exploit information about the system's current state to sustain production.
- **Populations:** Birth and death rates respond to age structure, resource availability, and dis-

ease dynamics— demographic feedback from the population's own state to its generative rate.

- **Firms:** Revenue generation depends on pricing, capital allocation, and management systems that encode a model of the firm's own cost structure and competitive position.

- **Software projects:** Commit activity is sustained by issue-tracking, code review, and maintenance systems that feed current project health back into developer contributions.

These feedback loops are *functional proxies for a self-model*: they do not implement PSC-grade semantic self-representation, but they do perform state-readout, information processing, and actuation—the three operations whose irreversibility cost is captured by $\lambda_\Psi E_\Psi$. Under this interpretation, the Reflexive Landauer Bound generalises to a *Feedback Regulation Bound*: the minimum thermodynamic overhead any self-regulating system must sustain beyond its operational generation/drain, regardless of whether it carries a formal semantic self-model. The $\Lambda/2 \approx 0.13$ margin is then the minimum fraction of generation that must be reserved for regulatory feedback infrastructure in any system that sustains itself.

Limitation. This interpretation does not make the cross-domain results derivations from first principles. The connection between the friction of state-dependent feedback loops and the formal $\lambda_\Psi E_\Psi$ term requires additional theoretical development to be made precise. The empirical results are consistent with the interpretation but do not prove it. They are offered as evidence that the IPT identifies a genuine structural feature of self-maintaining systems, not merely a property of PSC reflexive systems in the strict MFRR sense.

7 Falsification and Open Questions

Lean formalization status. The IPT formula and its three proof obligations have been machine-checked in `ugp-lean` (zero sorry):

- **A1 $\mathbf{A}_{\text{Lean}}$**: $|\psi| = 1/\varphi$ (PSC contraction rate equals the inverse golden ratio), certified as `abs_psi_eq_inv_phi` from the Fibonacci renormalization spectrum [20].
- **A2 $\mathbf{A}_{\text{Lean}}$**: $H(U(1)) = \ln(2\pi)$ (adjudication entropy of the PSC-selected $U(1)$ group equals $\ln(2\pi)$), certified as `A2_adjudication_entropy` + `entropy_formula_U1`.
- **A3 $\mathbf{A}_{\text{Lean}}$**: The PSC forward/backward symmetry gives a 1/2 overhead split, certified as `A3_forward_backward_split`.
- **IPT theorem $\mathbf{A}_{\text{Lean}}$**: $\rho_{\text{crit}} = 1 + \ln\varphi/(2\ln 2\pi)$ as `IPT_theorem`, module `UgpPhysicsLean.IPT.InformationProfitThreshold` (zero sorry).

The Profit–Curvature Equivalence ($G/D = \exp(\Lambda \int R_F \, dV)$) remains computationally certified (MFRR E32, 0.04% error) but not yet Lean-proved. The core IPT threshold is now $\mathbf{A}_{\text{Lean}}$.

Falsification criterion. The IPT derivation is falsifiable at two levels:

(F1) *Theoretical falsification*: A proof that the PSC fixed-point iteration does not require a golden-ratio convergence surplus—i.e., that the leading-order coefficient of δ_{PSC} in Lemma 3.1 is not $\ln\varphi/(2\ln 2\pi)$ but some other value—would revise the threshold. Such a proof would require exhibiting a PSC-satisfying reflexive system whose generative overhead has a different scaling.

(F2) *Computational falsification*: A computational experiment satisfying all MFRR modeling assumptions (PSC, Reflexive Landauer Bound, Ψ-field dynamics) but exhibiting sustained coherence with $\rho \le$ IPT, or coherence collapse with $\rho >$ IPT, over a sufficiently long run, would constitute evidence against the threshold.

Pre-registered prospective prediction. The population dynamics result identifies a testable prospective prediction: countries with birth/death ratio in $[1.0, 1.13)$ as of 2022–2025 will show near-zero or negative population growth over 2025–2030, while countries above 1.13 will show positive growth. This prediction is derived from theory (not fitted to the 2010–2022 data). *Action item*: archive this prediction on OSF or AsPredicted.org before 2030 World Bank data is released, to enable a prospective confirmatory test.

Open quantitative questions.

- **The value of λ_Ψ**: The coherence coupling constant λ_Ψ is not fully free: the MFRR Profit–Curvature Equivalence [13] establishes $G/D = \exp(\Lambda \int R_F \, dV)$, which constrains λ_Ψ through the requirement of consistency between the Reflexive Landauer Bound and the information-geometric threshold. Specifically, at the critical profit ratio (Gen/Drain $= 1 + \Lambda/2$) the Landauer and curvature expressions must agree, giving $\lambda_\Psi = k_B T(\Lambda/2)/\langle E_\Psi \rangle_c$ where $\langle E_\Psi \rangle_c$ is the coherence-field energy at criticality. This makes λ_Ψ *system-dependent but geometrically determined*: it is not a free universal constant but a ratio controlled by the geometry and temperature of the specific system. Deriving it for canonical PSC systems (e.g. the JT-gravity toy model of Paper 3 in this series [11]) is a concrete near-term goal.

- **Finite-size corrections**: The IPT derivation assumes infinite-time averages. Finite-time corrections to IPT for systems of bounded duration are not yet computed.

- **Multi-agent and network generalizations**: The derivation applies to a single reflexive system. Generalization to networks of interacting reflexive systems, where generation and drain are distributed across nodes, requires new analysis.

- **Stochastic regimes**: The derivation assumes deterministic generation and drain rates. How the threshold shifts under stochastic fluctuations (beyond the noise treatment in TE1.H) is an open question.

Cross-domain echoes. The IPT derivation is conditional on structural premises A1–A3 applying to the domain in question. For domains where these premises hold only approximately, or where the natural Gen/Drain observable differs from the economic or ecological ratio used above, the threshold may manifest differently or require renormalization (cf. Section 5.1). We note three observations from independent research programs as directional evidence; none constitutes a confirmed IPT result.

- **Nuclear magic numbers [B-bridge]:** In companion work [10], all seven nuclear magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$ are derived from the Nilsson shell model using GTE-predicted pion parameters (f_π, m_π), with empirical spin-orbit coupling $\kappa \approx 0.050$.

  A targeted Gen/Drain analysis [10] identifies the correct normalization: defining $\kappa_{\min}(N)$ as the minimum spin-orbit coupling for magic number N to appear as a shell gap $> 0.3\,\hbar\omega_0$, the ratio

  $$\begin{aligned} \kappa_{\mathrm{emp}}/\kappa_{\min}(N{=}50) &= 0.050/0.0435 \\ &= 1.149 \approx \mathrm{IPT} = 1.1309 \end{aligned}$$

  matches to 1.6%. The choice of $N = 50$ is physically motivated: the $1g_{9/2}$ shell closure is the *first robust spin-orbit-only magic number* beyond $N = 20$ (requiring no tensor-force correction, unlike $N = 28$). Among all magic numbers, only $N = 50$ matches the target $\kappa_{\mathrm{emp}}/\mathrm{IPT} = 0.044$; all others differ by 10–40× more (`nuclear_ipt_analysis.py`).

  The physical interpretation follows the P15 accounting reconciliation: $\kappa_{\mathrm{emp}} = \mathrm{IPT} \times \kappa_{\min}(50)$ means the empirical coupling carries the structural overhead $\Lambda/2 = 0.131$ above the viability floor. The nuclear system operates at exactly the IPT threshold for the canonical shell closure, consistent with the P15 buffer-zone structure (nuclei in $[\kappa_{\min}(50), \kappa_{\mathrm{emp}}) \equiv$ viability buffer; nuclei above κ_{emp} have robust shells).

  Claim grade: [B-bridge], 1.6% match. The N=50 normalization is physically motivated but not axiomatically derived from GTE.

- **Genetic code and prebiotic chemistry [B-bridge]:** The standard genetic code is the unique survivor of eight simultaneous biological viability criteria [19]. A supporting Gen/Drain analysis of prebiotic autocatalytic chemistry (using published rate constants for amino acid formation in lipid vesicles [7] and template-directed ligation [6]) finds: the minimum total enhancement factor for a self-sustaining prebiotic network to cross from non-viable to viable equals exactly IPT $= 1.1309\times$. The transition occurs when the combined vesicle compartmentalization and template-catalysis enhancement equals IPT; below this threshold the network is drain-dominated, above it, self-sustaining. The result confirms the P15 accounting structure: bare prebiotic chemistry ($\rho \approx 1.0$) operates in the buffer zone $[1.0, \mathrm{IPT})$; the IPT threshold marks the genuine viability boundary. **Claim grade:** [B-bridge], toy model with real rate constants.

- **Consciousness [C-speculative]:** The no-consciousness theorem for Turing-computable systems [15] (NEMS Paper 93) establishes a *structural* boundary for consciousness, not a Gen/Drain threshold. Whether integrated information theory (IIT) [24] admits a quantitative threshold $\Phi^* > 0$ related to IPT is an open research question noted here for completeness.
- **Organizational selection [$\mathbf{B^-}$]:** A survivorship-bias-corrected panel of U.S. public equity (SEC EDGAR XBRL [27]) reveals a **three-regime failure cascade**: defining G/D = ROIC/WACC, the annual failure rates are 6.81% (Destruction zone), 1.11% (Watch zone, $0 \leq G/D <$ IPT), and 0.22% (Safe zone, $G/D \geq$ IPT). Model selection confirms IPT as a **genuine second structural boundary**: ΔBIC $= -92.19$ (fifteen times the strong-evidence threshold). An inception-cohort prospective design (N = 1,158) eliminates the survivor-selection confound: Watch Zone entrants face HR $= 5.01\times$ the bankruptcy hazard of Safe Zone entrants (Cox $p = 0.0003$), satisfying the pre-specified [B$^-$] gate.

  A subsequent second-moment analysis (N = 12,741 company-years) confirms the **variance phase transition**: Watch Zone return standard deviation ($\sigma = 2.33$) is $1.79\times$ larger than Safe Zone ($\sigma = 1.30$; Levene $p = 0.046$); bimodality coefficient BC(Watch) $= 0.907 \gg 0.555$ threshold (Hartigan's dip test $p \approx 0$); escape probabilities $P_{\mathrm{W\to S}} = 0.12$ and $P_{\mathrm{W\to D}} = 0.14$ (near-symmetric, consistent with critical-boundary dynamics). This mirrors the *diverging susceptibility* $\chi \to \infty$ at the Ising critical point: at the phase boundary, fluctuations are maximal and outcomes are least predictable.

  The combined first-moment (HR $= 5.01\times$, ΔBIC $= -92.19$) and second-moment (BC = 0.907, Levene $p = 0.046$) evidence constitutes a thermodynamically coherent picture of IPT as a genuine organizational phase boundary across seven independent tests. **Claim grade: $\mathbf{B^-}$** — mean-field and variance phase-transition signatures both confirmed; replication in non-US equity universe required for **B**.

Caveat: These observations are directional only. None has been tested against the accounting reconciliation of Section 5.1, and the claims grades are [B-bridge] and [C-speculative] respectively. They are reported here to motivate future work, not as confirmed cross-domain validations of IPT.

8 Conclusion

We have derived the Information Profit Threshold $\mathrm{IPT} = 1+\ln\varphi/(2\ln 2\pi) \approx 1.1309$ from two structural inputs of the MFRR framework: the Perfect Self-Containment (PSC) requirement and the Reflexive Landauer Bound. The derivation identifies φ and 2π as natural scales arising from PSC fixed-point convergence and phase-space periodicity, respectively; the threshold is not an empirical fit but a derived consequence of these structures.

Three computational experiments—the TE1.H 2D field simulator, the TE2.1 evolutionary agent suite, and the E4 energy-model consistency check—provide consistent internal validation: coherence grows only in the supercritical regime ($\rho >$ IPT), evolutionary survivors settle above the threshold, and the Reflexive Landauer energy model is confirmed non-degenerate across 50 randomized configurations (all margins strictly positive, min margin 0.018). We emphasize that these are simulation results within the MFRR framework; physical or biological experiments remain future work.

The IPT provides a concrete, falsifiable criterion distinguishing parameter regimes that can sustain reflexive coherence from those that cannot, and it opens a quantitative research program connecting the thermodynamics of self-reference to the information economics of self-maintaining systems.

All three structural premises (A1: $|\psi| = 1/\varphi$; A2: $H(U(1)) = \ln 2\pi$; A3: 1/2 forward/backward split) and the IPT theorem itself ($\rho_{\mathrm{crit}} = 1 + \ln\varphi/(2\ln 2\pi)$) are now machine-checked in Lean 4 with zero `sorry` in module `UgpPhysicsLean.IPT.InformationProfitThreshold` (Theorem `IPT_theorem`, `ugp-physics-lean` [23]). The claim type is therefore $\mathbf{A}_{\mathrm{Lean}}$ relative to the formal PSC/Fibonacci/$U(1)$-entropy architecture: given the three structural premises A1–A3, the threshold value is a necessary logical consequence, machine-verified. The remaining scientific question is whether A1–A3 correctly abstract the physics of self-maintaining systems in each empirical domain; the derivation does not assert that biological or economic systems satisfy PSC axioms directly, only that the IPT threshold is the correct structural consequence when they do. This is still a major upgrade: the threshold is no longer a computational approximation but an analytic necessity within the formal model.

Remark on charged-fermion mass calibration. Whether the Reflexive Landauer Bound can directly set the charged-fermion mass calibration scale E_{base} is analysed in the companion paper [9]. The analysis finds a systematic anti-correlation: the MFRR Landauer bound predicts larger E_{base} precisely for the fermions where the UCL calibration gives smaller values. This constitutes an *anti-correlation theorem*: the Reflexive Landauer Bound is structurally ruled out as a direct mass-generation mechanism for Dirac (charged) fermions. The mechanism remains viable for Majorana (right-handed neutrino) mass generation, where the lepton-number-violating transition is irreversible — a condition that lies within the MFRR domain — and is an active research target.

Companion papers. A systematic cross-domain survey of the IPT threshold across eight independent domains is presented in [14], including organisational survival, microbial metabolism, natural language, and conformal field theory universality classes. A candidate theoretical mechanism deriving IPT as the efficiency ratio at the fixed point of a self-referential renormalization group is developed in [18].

A Derivation Details

A.1 Notation and setup

Let $\mathcal{S}$ be a reflexive system satisfying PSC with self-model field $\Psi : \mathcal{U} \to \mathbb{R}$ on domain $\mathcal{U} \subseteq \mathbb{R}^d$. Fix a temperature T and coherence coupling $\lambda_\Psi > 0$. Define:

$$E_\Psi[\Psi] := \int_{\mathcal{U}} \left(\alpha_1 \Psi^2 + \alpha_2 |\nabla \Psi|^2\right) dV,$$

with $\alpha_1, \alpha_2 > 0$ fixed. By Theorem 2.3, each PT event costs at least $\Delta E_{\text{PT}} \geq k_B T \ln n + \lambda_\Psi E_\Psi$.

A.2 Derivation of Lemma 3.1

Step 1: PSC as a fixed-point iteration. The PSC requirement demands that the map

$$\mathcal{F}_x : \Psi \mapsto \Psi' \quad \text{satisfies} \quad \Psi^* = \mathcal{F}_x(\Psi^*)$$

at every observed configuration x. For $\mathcal{F}_x$ to converge from an arbitrary initial $\Psi^{(0)}$ in the minimum number of steps, the contraction constant $\kappa = \|\mathcal{F}_x\|_{\text{Lip}}$ should be minimized subject to the constraint that $\mathcal{F}_x$ still represents a valid self-model update. The unique positive κ^* minimizing convergence time while satisfying $\kappa^{*2} + \kappa^* = 1$ is $\kappa^* = 1/\varphi$, where φ is the golden ratio. This follows from the identity $1/\varphi = \varphi - 1$ and the fact that the two-step recursion $\Psi^{(t+1)} = \mathcal{F}_x(\mathcal{F}_x(\Psi^{(t)}))$ has optimal contraction when the single-step contraction is $1/\varphi$.

Step 2: Energy overhead from convergence. The rate of PSC convergence at contraction $\kappa^* = 1/\varphi$ implies that the system performs work at rate proportional to $-\ln \kappa^* = \ln \varphi$ per step to contract the self-model update. This work is *additional* to the bare Landauer cost and must be supplied by the generative budget.

Step 3: Phase-space normalization. At each PT step, n degenerate states are resolved with angular separation $2\pi/n$ on the phase circle $[0, 2\pi)$. The convergence overhead rate is $\ln \varphi$ (Step 2); this must be normalized to a dimensionless fraction of the baseline PT energy scale. The baseline angular information content is $\ln n$ per step; integrated over the full phase circle of circumference 2π, the natural normalization unit is $\ln(2\pi)$. Forming the ratio:

$$\delta_{\text{PSC}} = \frac{\text{(conv. overhead rate)}}{\text{(angular info. norm.)}} = \frac{\ln \varphi}{\ln 2\pi}.$$

Numerically: $\ln \varphi \approx 0.4812$, $\ln(2\pi) \approx 1.8379$, and their ratio $\Lambda \approx 0.2618$.

Step 4: The 1/2 prefactor. The PSC consistency check operates on both the forward (generative) and the backward (verification) pass of the adjudication step. Only the forward pass contributes to the generative overhead; the backward pass is charged to the drain. Hence the net surplus attributed to the generative side is $\delta_{\text{PSC}}/2$.

Step 5: Threshold. The system is viable with PSC closure if and only if $G/D > 1 + \delta_{\text{PSC}}/2$, which gives Equation (3). □

B Simulation Methodology

B.1 TE1.H

The TE1.H simulator (`information_profit_simulation.py`) models a 64×64 2D real-valued field $f(x, y, t)$. At each step:

- **Generation:** $f \mathrel{+}= A_{\text{gen}} \sin(2\pi kx/L)\cos(2\pi ky/L)$, where A_{gen} is the generation amplitude and k the spatial frequency.
- **Drain:** $f \mathrel{\times}= (1 - r_{\text{drain}})$, where r_{drain} is the decay rate per step.
- **Levin noise:** $f \mathrel{+}= \xi_t$ where $\xi_t \sim \text{Uniform}[-A_{\text{noise}}, A_{\text{noise}}]$ independently at each grid point.
- **Coherence measurement:** $\mathcal{C}_t = 1 - |\text{gzip}(f_t)|/|f_t|$ (compressed size / raw size).

Parameters: Unprofitable ($A_{\text{gen}} = 0.07$, $r_{\text{drain}} = 0.08$, $A_{\text{noise}} = 0.03$); Profitable ($A_{\text{gen}} = 0.14$, $r_{\text{drain}} = 0.08$, $A_{\text{noise}} = 0.02$); High Noise ($A_{\text{gen}} = 0.14$, $r_{\text{drain}} = 0.08$, $A_{\text{noise}} = 0.12$). Each run: 400 steps, single seed.

B.2 TE2.1

The TE2.1 evolutionary genesis experiment (`MFRR_Evolutionary_Genesis.py`) evolves agents with small neural networks (weights $\in \mathbb{R}^{d_{\text{net}}}$) in a 2D discrete resource grid. Fitness is determined by survival time, which correlates with maintaining $\rho > 1$. The profit ratio is computed as the ratio of cumulative generation events (resource absorption) to cumulative drain events (metabolic cost + resource decay). The sweep experiment (`MFRR_Evolutionary_Sweep.py`) varies metabolic cost $c \in \{0.0008, 0.001, 0.0012, 0.0015\}$ and decay rate $d \in \{0.004, 0.0043, 0.0046, 0.0049, 0.0052, 0.0055\}$, 3 random seeds per grid point, 40 generations, 600 frames per run.

B.3 E4

The E4 consistency check (`E4_reflexive_landauer_check.py`) generates 50 independent random configurations. For each: a smooth 32×32 coherence field is generated (Gaussian filtering, $\sigma = 2.0$); parameters $n \in \{2, \ldots, 8\}$, $T \in [0.5, 1.0]$, $\lambda_\Psi \in [0.5, 2.0]$, $\alpha_1, \alpha_2 \in [0.5, 1.5]$ are drawn uniformly; a random difficulty $\epsilon \in [0, 1]$ is drawn; and

$$\begin{aligned} \Delta E_{\text{PT}} &= k_B T \ln n + (1+\epsilon)\,\lambda_\Psi E_\Psi, \\ \text{RHS} &= k_B T \ln n + \lambda_\Psi E_\Psi, \end{aligned}$$

are computed. The test verifies $\Delta E_{\text{PT}} \geq \text{RHS}$ (equivalently, $\epsilon \geq 0$, which holds by construction). The informative output is the margin $\epsilon\,\lambda_\Psi E_\Psi$, which is bounded away from zero across all configurations (min margin 0.018).

References

[1] Charles H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973.

[2] Qianyang Chen and Mikhail Prokopenko. Why collective behaviours self-organise to criticality: A primer on information-theoretic and thermodynamic utility measures. *Entropy*, 27(6):641, 2025.

[3] Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.

[4] Grégoire Nicolis and Ilya Prigogine. *Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order Through Fluctuations.* Wiley-Interscience, New York, 1977.

[5] Phillip Norfleet. Private communication, 2025.

[6] Leslie E. Orgel. Molecular replication. *Nature*, 358:203–209, 1992. Template-directed ligation kinetics in the RNA world.

[7] Bernd M. Rode. Peptides and the origin of life. *Peptides*, 20:773–786, 1999. Salt-induced peptide formation kinetics; rate constants for amino acid condensation in prebiotic conditions.

[8] David J. Schwab, Ilya Nemenman, and Pankaj Mehta. Zipf's law and criticality in multivariate data without fine-tuning. *Physical Review Letters*, 113:068102, 2014. arXiv:1310.0448. Shows that systems with unobserved latent variables generically produce Zipf-distributed marginal statistics without fine-tuning, placing them at criticality (E = S).

[9] N. Spivack. A deterministic number-theoretic framework for the standard model parameter spectrum. Zenodo, 2026. Paper P01 in the UGP Physics series. `https://doi.org/10.5281/zenodo.20168787`.

[10] Nova Spivack. A unified, first-principles theory of the atomic nucleus from a deterministic, gte-only framework. Zenodo, 2025. https://doi.org/10.5281/zenodo.20170082. Applies the GTE framework to nuclear binding energies and stability.

[11] Nova Spivack. Black hole unitarity via reflexive unitarity. UGP Physics Programme, Paper 16, 2026. https://doi.org/10.5281/zenodo.20170110. Models Hawking evaporation as a GKSL master equation on a JT-like PSC universe; Stinespring dilation of the evaporation channel confirms unitary information recovery (TE2.4).

[12] Nova Spivack. A formal theory of transputation. NEMS/PSC Suite, Paper 76, 2026. https://doi.org/10.5281/zenodo.19429882. Lean: `Transputation.Theorems.ForcedAdjudication` (transputation-lean, zero sorry).

[13] Nova Spivack. From NEMS to MFRR: A machine-checked bridge between semantic closure and reflexive reality. NEMS/PSC Suite, Paper 08, 2026. https://doi.org/10.5281/zenodo.19429729. Lean-verified bridge theorems and diagonal barrier; zero custom axioms.

[14] Nova Spivack. A general theory of selection: Two-stage sieves, the information profit threshold, and cyclotomic substrates. Zenodo, 2026. Paper P26 in the UGP Physics series. https://doi.org/10.5281/zenodo.20171011.

[15] Nova Spivack. The NEMS program: No external model selection — machine-checked suite (papers 00–92). Zenodo program hub, 2026. https://doi.org/10.5281/zenodo.19487299. Concept DOI (always latest): https://zenodo.org/records/19487299. 93 papers, zero-sorry Lean 4. Abstracts: https://novaspivack.github.io/research/abstracts/.

[16] Nova Spivack. PSC-optimality of the standard model in 4D gauge theory space. NEMS/PSC Suite, Paper 05, 2026. https://doi.org/10.5281/zenodo.19429721.

[17] Nova Spivack. Reflexive reality: A survey for physicists. NEMS/PSC Suite, Paper 88, 2026. https://doi.org/10.5281/zenodo.19429908.

[18] Nova Spivack. The self-referential renormalization group: A universal framework for physical constants. Zenodo, 2026. https://doi.org/10.5281/zenodo.20170859. Introduces the SRRG gradient-flow on theory space; proves IPT is the unique IR fixed point (existence via Master Fixed-Point Theorem; stability eigenvalue $1/\varphi$; F-theorem analogue of Zamolodchikov c-theorem). Derives $\theta_{\rm QCD} = 0$, $N_{\rm gen} = 3$, and the SM gauge structure (conditional theorem) from the SRRG fixed-point constraint. Machine-checked in `srrg-lean` (zero sorry, all owned modules).

[19] Nova Spivack. Structural admissibility and biological viability uniquely select the standard genetic code. Zenodo, 2026. Paper P25 in the UGP Physics series. https://doi.org/10.5281/zenodo.20170152.

[20] Nova Spivack. ugp-lean: Lean 4 formalization of UGP and GTE. https://github.com/novaspivack/ugp-lean, 2026. https://doi.org/10.5281/zenodo.19554700. More than 400 modules across seventeen architectural layers, including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3-Newton flow, binary cascade, physical mass spectrum), BraidAtlas (composite triples, chirality squaring, charge derivation), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. `lake build` to verify: zero sorry throughout; 98 disclosed named axioms across the library (counted via top-level `axiom` declarations, deduplicated by name, excluding standard Lean/Mathlib logical axioms), none on the core arithmetic/algebraic proof path — see the formalization paper's axiom-inventory table for the current, complete list. Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

[21] Nova Spivack. ugp-physics: Companion code and papers for the UGP physics series. https://github.com/novaspivack/ugp-physics, 2026. Includes `computational_concordance/` tables, basin JSON exports, and capstone LaTeX.

[22] Nova Spivack. ugp-physics-lean: Lean 4 formalization of IPT, GXT, and SRRG physics modules. https://github.com/novaspivack/ugp-physics-lean, 2026. Lean 4

library containing: GXT.H9SelfConsistency (ipt_self_consistent, zero sorry), GXT.GoldenRatioFixedPoint (golden_ratio_fixed_point_unique, zero sorry), GXT.U1DirectProof and GXT.LieExpSurjective (Circle topology, zero sorry for Circle; one sorry for abstract Lie group minimality pending LieGroup.exp in Mathlib), IPT.InformationProfitThreshold (entropy_formula_U1, A2_adjudication_entropy, IPT_theorem, all zero sorry). Zenodo concept DOI: 10.5281/zenodo.20171560.

[23] Nova Spivack. ugp-physics-lean: Lean 4 formalization of IPT, GXT, and SRRG physics modules. https://github.com/novaspivack/ugp-physics-lean, 2026. See SpivackUGPPhysicsLean. Zenodo concept DOI: 10.5281/zenodo.20171560.

[24] Giulio Tononi. Consciousness as integrated information: a provisional manifesto. *The Biological Bulletin*, 215(3):216–242, 2008.

[25] UCI Machine Learning Repository. Polish companies bankruptcy data. https://archive.ics.uci.edu/dataset/365/, 2016.

[26] U.S. Bureau of Labor Statistics. Business employment dynamics: Establishment births and deaths. https://www.bls.gov/bdm/, 2024. Annual private-sector establishment birth and death counts, 1994–2022. BED flat file download.

[27] U.S. Securities and Exchange Commission. EDGAR full-text search and XBRL financial data. https://efts.sec.gov/LATEST/search-index?q=%22XBRL%22&dateRange=custom, 2024. SEC EDGAR inline XBRL financial filings database, including historical filings from delisted companies; used for survivorship-bias-corrected bankruptcy analysis. Accessed 2026.

[28] World Bank. World development indicators: Crude birth rate and crude death rate. https://data.worldbank.org/indicator/, 2024. Indicators SP.DYN.CBRT.IN (crude birth rate per 1,000) and SP.DYN.CDRT.IN (crude death rate per 1,000). Annual data 1960–2023 for 261 countries/territories. Accessed 2026.

[29] Maciej Zięba, Sebastian M. Tomczak, and Jakub M. Tomczak. Ensemble boosted trees with synthetic features generation in application to bankruptcy prediction. *Expert Systems with Applications*, 58:93–101, 2016. Source of the UCI Polish Companies Bankruptcy Dataset; 10,503 Polish manufacturing companies, 2000–2013, 64 financial ratios including Attr58 = total costs/total sales.

Acknowledgments

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.